PGCP-AC AUGUST 26

Database Assignment No. 3

1. Write a query to calculate the total salary of all employees.

```
mysql> SELECT SUM(Salary) AS TotalSalary FROM Emp;
+-------------+
| TotalSalary |
+-------------+
|   375000.00 |
+-------------+
1 row in set
```

2. Find the average salary of employees in each department using GROUP BY.

```
mysql> SELECT DeptID, AVG(Salary) AS AverageSalary
    -> FROM Emp
    -> GROUP BY DeptID;
+--------+---------------+
| DeptID | AverageSalary |
+--------+---------------+
|     10 |      67500.00 |
|     20 |      69000.00 |
|     30 |      52000.00 |
+--------+---------------+
3 rows in set
```

3. Count the total number of employees in each department.

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Emp
    -> GROUP BY DeptID;
+--------+----------------+
| DeptID | TotalEmployees |
+--------+----------------+
|     10 |              2 |
|     20 |              2 |
|     30 |              1 |
+--------+----------------+
3 rows in set
```

4. Display departments having more than 5 employees using HAVING.

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Emp
    -> GROUP BY DeptID
    -> HAVING COUNT(*) > 5;
Empty set
```

5. List distinct department locations from the Dept table.

```
mysql> SELECT DISTINCT Location FROM Dept;
+----------+
| Location |
+----------+
| Mumbai   |
| Pune     |
| Delhi    |
+----------+
3 rows in set
```

6(a). Display the highest salary among all employees.

```
mysql> SELECT MAX(Salary) AS HighestSalary FROM Emp;
+---------------+
| HighestSalary |
+---------------+
|      90000.00 |
+---------------+
1 row in set
```

6(b). Display the total number of employees in the table.

```
mysql> SELECT COUNT(*) AS TotalEmployees FROM Emp;
+----------------+
| TotalEmployees |
+----------------+
|              5 |
+----------------+
1 row in set
```

7. Find employees whose name starts with 'A' using LIKE.

```
mysql> SELECT * FROM Emp WHERE EmpName LIKE 'A%';
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   104 | Amit    | 52000.00 |     30 | 2025-08-01  | Delhi  |
+-------+---------+----------+--------+-------------+--------+
1 row in set
```

8. Find employees whose name ends with 'n' using LIKE.

```
mysql> SELECT * FROM Emp WHERE EmpName LIKE '%n';
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   103 | Rohan   | 75000.00 |     10 | 2023-03-20  | Mumbai |
+-------+---------+----------+--------+-------------+--------+
1 row in set
```

9. Find employees whose name contains 'ra' using LIKE.

```
mysql> SELECT * FROM Emp WHERE EmpName LIKE '%ra%';
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   103 | Rohan   | 75000.00 |     10 | 2023-03-20  | Mumbai |
|   105 | Priya   | 90000.00 |     20 | 2022-11-12  | Pune   |
+-------+---------+----------+--------+-------------+--------+
2 rows in set
```

10. Display all employees sorted by salary in descending order.

```
mysql> SELECT * FROM Emp ORDER BY Salary DESC;
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   105 | Priya   | 90000.00 |     20 | 2022-11-12  | Pune   |
|   103 | Rohan   | 75000.00 |     10 | 2023-03-20  | Mumbai |
|   101 | Tejas   | 60000.00 |     10 | 2025-01-15  | Mumbai |
|   104 | Amit    | 52000.00 |     30 | 2025-08-01  | Delhi  |
|   102 | Khushi  | 48000.00 |     20 | 2024-06-10  | Pune   |
+-------+---------+----------+--------+-------------+--------+
5 rows in set
```

11. Sort employees by DeptID ascending and then Salary descending.

```
mysql> SELECT * FROM Emp ORDER BY DeptID ASC, Salary DESC;
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   101 | Tejas   | 60000.00 |     10 | 2025-01-15  | Mumbai |
|   103 | Rohan   | 75000.00 |     10 | 2023-03-20  | Mumbai |
|   105 | Priya   | 90000.00 |     20 | 2022-11-12  | Pune   |
|   102 | Khushi  | 48000.00 |     20 | 2024-06-10  | Pune   |
|   104 | Amit    | 52000.00 |     30 | 2025-08-01  | Delhi  |
+-------+---------+----------+--------+-------------+--------+
5 rows in set
```

12. Find employees whose salary is between 30,000 and 60,000.

```
mysql> SELECT * FROM Emp WHERE Salary BETWEEN 30000 AND 60000;
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   101 | Tejas   | 60000.00 |     10 | 2025-01-15  | Mumbai |
|   102 | Khushi  | 48000.00 |     20 | 2024-06-10  | Pune   |
|   104 | Amit    | 52000.00 |     30 | 2025-08-01  | Delhi  |
+-------+---------+----------+--------+-------------+--------+
3 rows in set
```

13. Display employees whose DeptID is in (10, 20, 30).

```
mysql> SELECT * FROM Emp WHERE DeptID IN (10, 20, 30);
+-------+---------+----------+--------+-------------+--------+
| EmpID | EmpName | Salary   | DeptID | JoiningDate | City   |
+-------+---------+----------+--------+-------------+--------+
|   101 | Tejas   | 60000.00 |     10 | 2025-01-15  | Mumbai |
|   102 | Khushi  | 48000.00 |     20 | 2024-06-10  | Pune   |
|   103 | Rohan   | 75000.00 |     10 | 2023-03-20  | Mumbai |
|   104 | Amit    | 52000.00 |     30 | 2025-08-01  | Delhi  |
|   105 | Priya   | 90000.00 |     20 | 2022-11-12  | Pune   |
+-------+---------+----------+--------+-------------+--------+
5 rows in set
```

14. Display the minimum salary of an employee.

```
mysql> SELECT MIN(Salary) AS MinimumSalary FROM Emp;
+---------------+
| MinimumSalary |
+---------------+
|      48000.00 |
+---------------+
1 row in set
```

15. Display employees whose JoiningDate is between 2020-01-01 and 2021-12-31.

```
mysql> SELECT * FROM Emp
    -> WHERE JoiningDate BETWEEN '2020-01-01' AND '2021-12-31';
Empty set
```

16. Display employees whose Salary is NULL.

```
mysql> SELECT * FROM Emp WHERE Salary IS NULL;
Empty set
```

17. Display employees whose Salary is NOT NULL.

```
mysql> SELECT * FROM Emp WHERE Salary IS NOT NULL;
5 rows in set
```

18. Calculate total salary per department, only where total salary is greater than 100000.

```
mysql> SELECT DeptID, SUM(Salary) AS TotalSalary
    -> FROM Emp
    -> GROUP BY DeptID
    -> HAVING SUM(Salary) > 100000;
+--------+-------------+
| DeptID | TotalSalary |
+--------+-------------+
|     10 |   135000.00 |
|     20 |   138000.00 |
+--------+-------------+
2 rows in set
```

19. Display all distinct employee names.

```
mysql> SELECT DISTINCT EmpName FROM Emp;
+---------+
| EmpName |
+---------+
| Tejas   |
| Khushi  |
| Rohan   |
| Amit    |
| Priya   |
+---------+
5 rows in set
```

20. Count the number of departments having the same location.

```
mysql> SELECT Location, COUNT(*) AS DepartmentCount
    -> FROM Dept
    -> GROUP BY Location
    -> HAVING COUNT(*) > 1;
Empty set
```